

Central Near-Vision (CN) Design Ortho-K Lens

R&D Test Report

Objective: Determine whether the CN-design lens delivers meaningfully better near vision than a standard Ortho-K lens, and identify the parameter sensitivities that drive the near-vision response.

1. Test Objectives

1. Verify whether the CN-design lens improves near vision compared to a standard Ortho-K lens.
2. Change only one parameter per iteration, and compare the four-way response: distance-with-lens, near-with-lens, distance-post-removal, near-post-removal.

2. Test Subject

Eye	Myopia Level	Spherical Equivalent	Dominant Eye
Right (OD)	Low myopia	≈ -2.00 D	Yes
Left (OS)	High myopia	≈ -5.50 D	No

Each eye was tuned independently, forming two parallel iteration tracks:

OD → Low-Myopia Track · OS → High-Myopia Track · 5 rounds each.

3. Method

Baseline = prior distance-only design parameters. CN lenses were fabricated; in each round one variable was changed (C factor / ADD / Power). Four outcomes were recorded per round:

- **Distance vision with lens on** (on-lens distance)
- **Near vision with lens on** (on-lens near)
- **Distance vision after removal** (post-removal distance)
- **Near vision after removal** (post-removal near)

4. OD Iteration Track — Low Myopia (≈ -2.00 D)

#	BC	Power	Dia.	ADD	C factor	Change	On-lens Dist.	On-lens Near	Post-removal Dist.	Post-removal Near
R1	44.25	-1.25	10.6	+1.5D	4.0	Baseline	Very good	Slightly better than prior MCOK	Very good	Baseline; near below expectation
R2	44.25	-1.25	10.6	+1.5D	6.0 ↑	C ↑	No effect	—	Better (prior MCOK over-corrected)	C↑ no negative impact on distance; near ~same as R1
R3	44.25	-1.25	10.6	+2.5D ↑	6.0	ADD ↑	Clear	Holding up	0.8 (next-day wear)	ADD↑ improved near (next-day wear)

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R4	44.25	-1.25	10.6	+2.5D	4.0 ↓	C ↓	Clear	Similar to R3	—	C 6.0 → 4.0: negligible difference
R5	44.25	-1.25	10.6	+2.5D	1.0 ↓↓	C ↓↓	Mediocre	Worse	—	C too low; near degrades

OD Summary

- Toggling C factor between 4.0 ↔ 6.0 produced little observable difference → **low myopia appears C-factor-insensitive within this range.**
- Pushing C factor to the extreme (1.0) clearly worsened near → **a lower bound exists.**
- Raising ADD from +1.5 D → +2.5 D improved near on next-day wear; usable at normal reading distance but not equivalent to +2.5 D reading glasses.
- Overall, near vision was **modestly better** than a standard Ortho-K lens, **but did not reach the target outcome.**

5. OS Iteration Track — High Myopia (≈ -5.50 D)

#	BC	Power	Dia.	ADD	C factor	Change	On-lens Dist.	On-lens Near	Post-removal Dist.	Post-removal Near
L1	44.00	-4.75	10.6	+2.5D	4.0	Baseline	0.6	Comfortable	0.6 (under-corrected)	Affected by sleep and wear duration
L2	44.00	-6.00 ↑↑	10.6	+2.5D	4.0	Power ↑↑	Blurry (small-ring)	Poor	1.2	Multi-focal sensation at near
L3	44.00	-5.25 ↓	10.6	+2.5D	4.0	Power ↓	Clear	Poor	0.8 (Continue 2 nights wearing)	Dist. clear after down-titration; near mediocre
L4	44.00	-5.25	10.6	+2.5D	6.0 ↑	C ↑	Clear	Clear, better than before	Good	C↑ improved both distance and near
L5	44.00	-5.25	10.6	+2.5D	10.0 ↑↑	C ↑↑	Worse	No improvement	—	C too high; dist. degrades, near no lift

OS Summary

- **Power path:** -4.75 → -6.00 (over-corrected, back off) → -5.25 (balance point).
- **C factor path:** 4.0 → 6.0 (improved) → 10.0 (degraded) → **high myopia is C-factor-sensitive with an apparent optimum near 6.0.**

6. Cross-Comparison

Dimension	OD — Low Myopia	OS — High Myopia
C factor sensitivity	Low — 4.0 ↔ 6.0 indistinguishable; only breaks at 1.0	High — 4.0 → 6.0 improves, 6.0 → 10.0 reverses
C factor optimum	≥ 4.0 sufficient; no need to push higher	~ 6.0
ADD effect	Core variable; +2.5 D improves near	Not isolated (held at +2.5 D throughout)

Power tuning	Not adjusted (−1.25 stable)	Fine-tuning required; narrow window
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7. Overall Findings

(AI-generated synthesis — to be reviewed)

3. **ADD is the primary driver of near vision.** Effect was clearest on the low-myopia eye (R2 → R3).
4. **C factor sensitivity is power-dependent.** Low myopia tolerates a wide C-factor range (4.0–6.0 ≈ equivalent); high myopia shows a narrow optimum near 6.0, with 10.0 actively degrading distance.
5. **C factor also affects lens cosmesis** — the higher the C, the more visible the central ring (barely visible at 1.0 → prominent at 10.0).
6. **High myopia has a narrower tuning window** on both axes — both power over-correction (−6.00) and excessive C factor (10.0) required back-off.

8. Open Questions & Critical Notes

(For review before acting on the findings)

- **“Near did not reach expectation” (OD&OS) is a subjective endpoint.** Before the next cycle, consider defining a quantitative near-vision threshold (e.g., reading acuity at 40 cm, or a specific J-value) so “meets expectation” is falsifiable.
- **ADD was never isolated on the high-myopia track.** Whether high myopia responds to ADD the way low myopia does remains untested.

The human is the driver, and AI is the co-driver. @Clark